

数学物理方程(理科版)

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第四章 位势方程

§1. 基本解与Green函数

§1.1 基本解与Green公式

(5) $\Gamma(x, \xi)$ 的性质:

$$\Gamma(x, \xi) = \begin{cases} -\frac{1}{2\pi} \ln|x - \xi|, & n = 2, \\ \frac{1}{(n-2)\omega_n} \frac{1}{|x - \xi|^{n-2}}, & n \geq 3. \end{cases}$$

- ① $\Gamma(x, \xi) \in C^\infty(\mathbb{R}^n \setminus \{\xi\})$, $\Gamma(x, \xi) \in L_{loc}(\mathbb{R}^n)$;
- ② $\Delta \Gamma(x, \xi) = 0$, $x \in \mathbb{R}^n \setminus \{\xi\}$; $\lim_{x \rightarrow \xi} \Gamma(x, \xi) = +\infty$;
- ③ $\Gamma(x, \xi)$ 是Laplace方程的基本解, 即

$$-\Delta \Gamma(x, \xi) = \delta(x - \xi), \quad x, \xi \in \mathbb{R}^n$$

第1条和第2条可以直接计算验证, 下面来证明第3条.

第3条的证明： 只证 $n = 2$ 的情况， $n \geq 3$ 的证明是类似。

要证： $-\Delta\Gamma(x, y; \xi, \eta) = \delta(x - \xi, y - \eta)$, $(x, y), (\xi, \eta) \in \mathbb{R}^2$

即 $\langle -\Delta\Gamma(x, y; \xi, \eta), \varphi(x, y) \rangle = \langle \delta(x - \xi, y - \eta), \varphi(x, y) \rangle = \varphi(\xi, \eta), \forall (\xi, \eta) \in \mathbb{R}^2, \forall \varphi \in \mathcal{D}(\mathbb{R}^2)$.

因为 $\Gamma(x, y; \xi, \eta) \in L_{loc}(\mathbb{R}^2)$. 等价于要证

$$(*) \quad \begin{cases} -\iint_{\mathbb{R}^2} \Gamma(x, y; \xi, \eta) \Delta\varphi(x, y) dx dy = \varphi(\xi, \eta), \\ \forall (\xi, \eta) \in \mathbb{R}^2, \forall \varphi \in \mathcal{D}(\mathbb{R}^2) \end{cases}$$

$\forall \varepsilon > 0$, 令

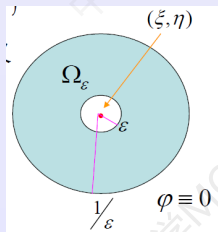
$$\Omega_\varepsilon = \{(x, y) \in \mathbb{R}^2 \mid \varepsilon^2 < (x - \xi)^2 + (y - \eta)^2 < \frac{1}{\varepsilon^2}\}$$

则 $\Gamma(x, y; \xi, \eta) \in C^\infty(\overline{\Omega_\varepsilon})$. $\forall \varphi \in \mathcal{D}(\mathbb{R}^2)$, 可取 ε 足够小, 使得当 $(x - \xi)^2 + (y - \eta)^2 \geq \frac{1}{\varepsilon^2}$ 时, $\varphi \equiv 0$. 在 Ω_ε 上对 Γ 和 φ 应用 Green 公式, 得

$$\iint_{\Omega_\varepsilon} (\Gamma \Delta \varphi - \varphi \Delta \Gamma) dx dy = \int_{\partial \Omega_\varepsilon} \left(\Gamma \frac{\partial \varphi}{\partial \mathbf{n}} - \varphi \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds.$$

注意到

$$\begin{cases} \Delta \Gamma = 0, (x, y) \in \Omega_\varepsilon \\ \varphi = 0, \frac{\partial \varphi}{\partial \mathbf{n}} = 0, \text{ 当 } (x - \xi)^2 + (y - \eta)^2 = \frac{1}{\varepsilon^2}. \end{cases}$$



就有

$$\iint_{\Omega_\varepsilon} \Gamma \Delta \varphi dx dy = \int_{\rho=\varepsilon} \left(\Gamma \frac{\partial \varphi}{\partial \mathbf{n}} - \varphi \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds,$$

这里 $\rho = \sqrt{(x - \xi)^2 + (y - \eta)^2}$. 所以

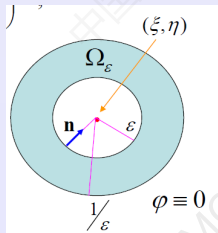
$$\iint_{\mathbb{R}^2} \Gamma \Delta \varphi dx dy = \lim_{\varepsilon \rightarrow 0} \iint_{\Omega_\varepsilon} \Gamma \Delta \varphi dx dy = \lim_{\varepsilon \rightarrow 0} \int_{\rho=\varepsilon} \left(\Gamma \frac{\partial \varphi}{\partial \mathbf{n}} - \varphi \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds.$$

注意到 $\mathbf{n}$ 是 Ω_ε 在 $\partial\Omega_\varepsilon$ 上的单位外法向量, 故在圆 $\rho = \varepsilon$ 上,

$$\frac{\partial}{\partial \mathbf{n}} = -\frac{\partial}{\partial \rho}.$$

而对 $n = 2$, Γ 表达式为

$$\Gamma = -\frac{1}{2\pi} \ln \sqrt{(x - \xi)^2 + (y - \eta)^2} = -\frac{1}{2\pi} \ln \rho.$$



所以在 $\rho = \varepsilon$ 上,

$$\begin{cases} \Gamma = -\frac{1}{2\pi} \ln \varepsilon, \\ \left. \frac{\partial \Gamma}{\partial \mathbf{n}} \right|_{\rho=\varepsilon} = -\left. \frac{\partial \Gamma}{\partial \rho} \right|_{\rho=\varepsilon} = -\frac{1}{2\pi} \left. \frac{\partial \ln \rho}{\partial \rho} \right|_{\rho=\varepsilon} = -\frac{1}{2\pi \varepsilon} \end{cases}$$

因此

$$\begin{aligned} \left| \int_{\rho=\varepsilon} \Gamma \frac{\partial \varphi}{\partial \mathbf{n}} ds \right| &= \frac{1}{2\pi} |\ln \varepsilon| \left| \int_{\rho=\varepsilon} \frac{\partial \varphi}{\partial \mathbf{n}} ds \right| \leq \frac{1}{2\pi} |\ln \varepsilon| \max_{\rho=\varepsilon} |\nabla \varphi| \left| \int_{\rho=\varepsilon} ds \right| \\ &= \frac{1}{2\pi} |\ln \varepsilon| \max_{\rho=\varepsilon} |\nabla \varphi| \cdot 2\pi \varepsilon \end{aligned}$$

故

$$\lim_{\varepsilon \rightarrow 0} \int_{\rho=\varepsilon} \Gamma \frac{\partial \varphi}{\partial \mathbf{n}} ds = 0;$$

而

$$\int_{\rho=\varepsilon} \varphi \frac{\partial \Gamma}{\partial \mathbf{n}} ds = -\frac{1}{2\pi\varepsilon} \int_{\rho=\varepsilon} \varphi ds.$$

由积分中值定理, 存在 (x_1, y_1) 满足 $(x_1 - \xi)^2 + (y_1 - \eta)^2 = \varepsilon^2$, 使得

$$-\frac{1}{2\pi\varepsilon} \int_{\rho=\varepsilon} \varphi ds = -\frac{\varphi(x_1, y_1)}{2\pi\varepsilon} \int_{\rho=\varepsilon} ds = -\varphi(x_1, y_1)$$

所以

$$\lim_{\varepsilon \rightarrow 0} \int_{\rho=\varepsilon} \varphi \frac{\partial \Gamma}{\partial \mathbf{n}} ds = -\lim_{\varepsilon \rightarrow 0} \varphi(x_1, y_1) = -\varphi(\xi, \eta)$$

于是

$$\iint_{\mathbb{R}^2} \Gamma \Delta \varphi dx dy = \lim_{\varepsilon \rightarrow 0} \int_{\rho=\varepsilon} \left(\Gamma \frac{\partial \varphi}{\partial \mathbf{n}} - \varphi \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds = -\varphi(\xi, \eta).$$

证毕.

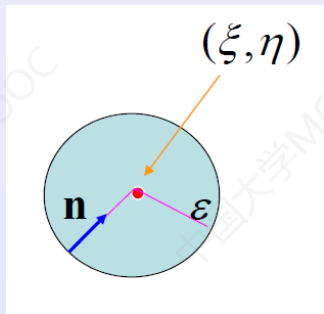
注：在上面的证明中，得到

$$\begin{cases} \lim_{\varepsilon \rightarrow 0} \int_{\rho=\varepsilon} \Gamma \frac{\partial \varphi}{\partial \mathbf{n}} ds = 0; \\ \lim_{\varepsilon \rightarrow 0} \int_{\rho=\varepsilon} \varphi \frac{\partial \Gamma}{\partial \mathbf{n}} ds = \varphi(\xi, \eta). \end{cases}$$

其中

$$\begin{cases} \rho = \sqrt{(x - \xi)^2 + (y - \eta)^2}, \\ \Gamma = \Gamma(x, y; \xi, \eta) \text{ 是基本解,} \\ \varphi = \varphi(x, y) \text{ 在 } (\xi, \eta) \text{ 的一个领域内有连续偏导数} \\ \mathbf{n} \text{ 在为圆 } \rho = \varepsilon \text{ 的} \text{单位内法向量}. \end{cases}$$

这两个结果我们后面还要用.



积分表示式:

定理1.3: 设 Ω 为 $\mathbb{R}^2$ 中的有界开区域, 其边界 $\partial\Omega$ 分片光滑.

$u \in C^2(\overline{\Omega})$, $\Gamma(x, y; \xi, \eta)$ 是Laplace方程的基本解. 则对 $\forall(\xi, \eta) \in \Omega$ 成立:

$$u(\xi, \eta) = - \iint_{\Omega} \Gamma \Delta u dx dy + \int_{\partial\Omega} \left(\Gamma \frac{\partial u}{\partial \mathbf{n}} - u \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds.$$

这里 $\mathbf{n} = (n_x, n_y)$ 是 $\partial\Omega$ 上的单位外法向量.

证明: 与前面证明类似. 令

$$\Omega_\varepsilon = \{(x, y) \in \Omega \mid \varepsilon^2 < (x - \xi)^2 + (y - \eta)^2\};$$

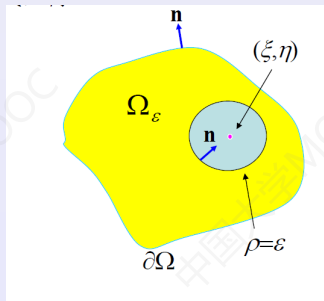
取 ε 足够小, 使

$$\left\{ (x, y) \in \mathbb{R}^2 \mid \rho \equiv \sqrt{(x - \xi)^2 + (y - \eta)^2} = \varepsilon \right\} \subset \Omega.$$

在 Ω_ε 上对 Γ 和 u 应用Green第二公式,得

$$\begin{aligned}\iint_{\Omega} \Gamma \Delta u dx dy &= \lim_{\varepsilon \rightarrow 0} \iint_{\Omega_\varepsilon} \Gamma \Delta u dx dy \\&= \lim_{\varepsilon \rightarrow 0} \iint_{\Omega_\varepsilon} (\Gamma \Delta u - u \Delta \Gamma) dx dy \\&= \lim_{\varepsilon \rightarrow 0} \int_{\partial \Omega_\varepsilon} \left(\Gamma \frac{\partial u}{\partial \mathbf{n}} - u \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds \\&= \int_{\partial \Omega} \left(\Gamma \frac{\partial u}{\partial \mathbf{n}} - u \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds + \lim_{\varepsilon \rightarrow 0} \int_{\rho=\varepsilon} \left(\Gamma \frac{\partial u}{\partial \mathbf{n}} - u \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds \\&= \int_{\partial \Omega} \left(\Gamma \frac{\partial u}{\partial \mathbf{n}} - u \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds + 0 - u(\xi, \eta).\end{aligned}$$

证毕.



注1: 定理1.3在维数大于2时也成立.

注2: 公式

$$u(\xi, \eta) = - \iint_{\Omega} \Gamma \Delta u dx dy + \int_{\partial\Omega} \left(\Gamma \frac{\partial u}{\partial \mathbf{n}} - u \frac{\partial \Gamma}{\partial \mathbf{n}} \right) ds, \quad \forall (\xi, \eta) \in \Omega,$$

说明了: u 在 Ω 中任一点的值可由其在 Ω 中的 Δu 以及在 $\partial\Omega$ 上的值 u 及 $\frac{\partial u}{\partial \mathbf{n}}$ 确定, 这为求解位势方程的边值问题提供了重要的思路.

同学们，再见！